

Adaptation Experiment 4: TS-IFA

1 Theoretical overview and goal

TS-IFA is a trainable adapter around a frozen forecast backbone. It forms four candidates: vanilla prediction $\hat{y}_0$, context-conditioned prediction $\hat{y}_c$, a residual-attention prediction $\hat{y}_r$, and a memory-attention prediction $\hat{y}_m$. Cross-attention reads retrieved examples and the final forecast is a horizon-wise convex mixture

$$\hat{y}_h = \sum_{b \in \{0, c, r, m\}} \pi_{b,h} \hat{y}_{b,h}, \quad \pi_{b,h} = \text{softmax}_b(a_{b,h}).$$

Residual attention uses neighbor backbone errors as values; memory attention uses neighbor futures. A third attention block produces mixture logits from query and neighbor look-backs. The horizon-specific four-way mixture is the highest-capacity final component and is the first target for simplification if T_2 overfitting persists.

Experimental goal. Learn a small, backbone-agnostic adaptation mechanism that can exploit retrieved forecasts, outcomes, and errors while remaining anchored to the reliable vanilla forecast.

2 Objective and initialization

With nMSE scaling, training minimizes

$$\mathcal{L} = \mathcal{L}_{\text{pred}} + \beta \|\hat{y} - \hat{y}_0\|_{\text{n}}^2 + \gamma (\|\Delta_r - (y - \hat{y}_0)\|_{\text{n}}^2 + \|\Delta_m - (y - \hat{y}_0)\|_{\text{n}}^2).$$

Correction heads are zero-initialized. Non-vanilla mixture logits start at -6 , so the initial adapter is close to the backbone and learns deviations only when supported by data.

3 Implementation details

Each attention block uses four heads of dimension 32, followed by a 128-unit GELU MLP. Inputs are instance-normalized on the query scale. AdamW uses learning rate 10^{-5} , weight decay 10^{-4} , gradient clipping 1, $\beta = \gamma = 10^{-2}$, batch size 256, and 10,000 epochs. Every epoch draws one random batch from T_1 . Full deterministic T_2 validation and interval logging occur every 1,000 optimizer steps; T_3 is evaluated only after training.

The trainer can restore the lowest-nMSE T_2 checkpoint before touching T_3 and can stop after a configurable number of non-improving validation evaluations. The launcher enables best-checkpoint restoration but leaves early stopping disabled by default for backward-compatible optimization length. Dropout and reduced attention/MLP dimensions should be screened on one pilot configuration before widening the sweep. A scalar/shared or low-rank final mixture is the next architecture ablation, not yet a default change.

The sweep covers Electricity/Solar, 168–24 and 672–168, raw/instance Euclidean retrieval, and $k \in \{1, 3, 10\}$. Implementation is under `src/adaptors/ts_ifa/`; the launcher is `src/slurm/run_ts_ifa.slurm`. Saved artifacts include the model, resolved configuration, full step history, interval-average train and T_2 validation plot, T_3 metrics, and branch predictions.

4 Interpretation

Report TS-IFA beside vanilla, context, residual, and memory branches. This distinguishes gains from better candidates from gains due to mixture learning. Selection must use T_2 ; the untouched T_3 period is the only final test. Attention weights are diagnostics and should not be interpreted as causal feature importance.